लोक सेवा आयोग,
राजपत्राङ्कित तृतीय श्रेणी, नेपाल इन्जिनियरिङ्ग सेवा, सिभिल समूह, जनरल उपसमूहको प्रतियोगितात्मक लिखित परीक्षा
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Section-A (25 Marks)

1. What are the design and construction problems of hill roads? What do you understand by obligatory points? What are the special considerations which need to be followed in the selection of alignments for roads in mountainous regions? [10]

Answer:

Hill road can be defined as the road that passes through the area with a cross slope of 25% or more. The hilly regions have extreme climatic condition, rugged topography and high altitude. The roads in these areas are affected by landslides, soil erosion and other problems.

Some of the major design and construction problems associated with hill roads are as follows:

i. **Topographical challenges:**
Highly broken relief, steep slope, deep gorges and large number of watercourses may unnecessarily increase the road length.

ii. **Geological and climatic variations:**
Great variation in the geological, hydrological and climatic conditions from one place to another.

iii. **Slope instability:**
Unstable slope after road construction triggers landslides, rockfalls.

iv. **Installation of various road structures:**
Hill roads require construction of various road structures such as retaining walls, parapet walls, breast walls, side drains, catch-water drains and others.

v. **Design of hairpin bends to gain height:**
Steep elevated area requires hairpin bends and sharp curves which restricts the vehicle speed and other geometric design parameters of hill road.

Obligatory points are those control points governing the highway alignment that tells if the alignment must pass through or should be avoided because of engineering, economic, social, or environmental considerations.

Some examples of the obligatory points through which the alignment should pass are:

i. Bridge site: The highway alignment can be changed to locate the bridge. It can be located where the river is straight and has permanent path and where the abutments and piers are strongly founded.

ii. Mountain: while the alignment passes through a mountain, the alternative is either to construct a tunnel or to go round the hills.
iii. Intermediate town: The alignment can be slightly deviated to connect an intermediate town nearby.

Some examples through which road alignment should not pass are as follows:
i. Religious places: these places have been protected by law from being acquired hence it must be avoided while aligning the highway.
ii. Costly structures: acquiring structures need heavy compensation which results in increase in the cost hence the alignment may be slightly deviated not to pass through that point.
iii. Lakes/ponds: the presence of lakes and ponds on the alignment requires the path to be deviated.
iv. Marshy lands, water logged areas etc.

When selecting the alignment of roads in mountainous (hilly) regions**,** special care is required due to steep slopes, unstable geology, and difficult terrain.

The special considerations which need to be followed in the selection of alignments for roads in mountainous regions are:

i. **Following Natural Contours**
   a. The road must be aligned as close as possible to the natural ground contours.
   b. The excessive cutting and filling must be minimized to reduce construction cost and slope instability.

ii. **Avoidance of Steep Gradients**
   a. The road gradient must be kept within permissible limits.
   b. Hairpin bends, switchbacks, or loops may be used where necessary to gain elevation gradually.

iii. **Selection of Stable Terrain**
   a. landslide-prone areas, rockfall zones, fault lines, and weak geological formations must be avoided while selecting the alignment.
   b. The stable rock and soil conditions must be chosen.

iv. **Minimization of Cross Drainage Works**
   a. The number of bridges, culverts, and retaining structures must be reduced by selecting suitable alignments.
   b. Cross streams may be provided at narrow and stable locations.

v. **Ensuring Proper Drainage**
   a. Efficient side drains, catch-water drains, and cross-drainage structures must be provided along the highway alignment.
   b. Water accumulation must be prevented that may weaken slopes.

vi. **Avoidance of Unstable Slopes**
   a. Stay away from areas susceptible to erosion, avalanches, and snow slides (where applicable).
   b. Adequate slope protection measures must be ensured.

vii. **Reduction in Earthwork**
   a. The cut and fill operations must be balanced to lower construction costs and environmental impacts.
   b. Deep cuttings and high embankments must be avoided whenever possible.

viii. **Provision of Adequate Sight Distance**
   a. Sufficient visibility at curves and bends must be ensured.
   b. Safe horizontal and vertical alignments should be designed.

ix. **Design of Safe Curves**
   a. Appropriate curve radii, transition curves, and superelevation must be designed and constructed.
   b. The pavements at sharp curves and hairpin bends should be widened in adequate amount.

x. **Consideration of Environmental and Social Impacts**

a. Forests, wildlife habitats, and environmentally sensitive areas must be considered before designing highway alignment.

b. The displacement of settlements and agricultural land must be minimized while selecting highway alignment.

2. **Describe briefly the methodology used in the construction of gravel roads. Differentiate between surface dressing treatment and otter seal construction. [10]**

Answer:

Gravel roads are unpaved surfaces constructed from quarried stone or streambed materials that are considered superior to earthen roads as they carry heavier traffic loads. This type of road can cater about 100 ton of pneumatic tired vehicle or 60 ton of iron-tired vehicles per lane.

The methodology used in the construction of gravel roads are as follows:

i. Gravel to be used for the construction is stacked along the sides of the proposed road.

ii. Location and preparation of subgrade:

- Compacted subgrade is prepared by using trench type of construction technology.
- Trench is formed to the desired depth of construction where the width of the trench is made equal to that of carriageway.
- The trench is brought to the desired grade and compaction of trench base is done to obtain hard-consolidated base for the gravel layer.

iii. Pavement construction:

- Crushed gravel aggregates are placed carefully in the trench to avoid segregation.
- Aggregates are spread with greater thickness at the center and less towards the edges to obtain the desired camber.
- The layer is rolled using smooth wheel rollers, from the edges to the centre with an overlap of at least half the width of the roller in the longitudinal direction.
- Camber is checked and corrected using camber board.

iv. Opening to traffic:

- Few days after the initial rolling and drying out the road is opened to traffic.
- Efficient drainage facilities should be provided.

The differences between surface dressing treatment and otta seal construction are given below:

Surface Dressing Treatment		Otta Seal Construction	
i.	A thin bituminous surface treatment in which bitumen is sprayed on the road surface and covered with a single-sized aggregate is known as surface dressing treatment.	i.	A bituminous surfacing using a relatively soft binder covered with well-graded aggregate, which is rolled into the binder to form a durable wearing course is known as otta seal construction.
ii.	Surface dressing treatment uses **uniform-sized (single-sized)** crushed stone aggregate.	ii.	Otta seal construction uses **well-graded aggregate** containing a range of particle sizes.
iii.	Generally, it uses **conventional bitumen or bitumen emulsion**.	iii.	It uses **soft bitumen or bitumen emulsion**, allowing better penetration into the aggregate.
iv.	Surface dressing is a thin layer, usually **10–20 mm**.	iv.	Otta seal is a thicker layer, usually **16–32 mm**.
v.	At first, binder is spread after which aggregates are spread and rolled with roller.	v.	At first, binder is sprayed after which graded aggregates are spread. Then, it is rolled repeatedly until aggregate is embedded.
vi.	Comparatively **less expensive than otta seal construction**.	vi.	Slightly **more expensive than surface dressing** initially but more economical over its service life.
vii.	It is suitable for **light to medium traffic** and has a shorter service life.	vii.	It is more durable and suitable for **low- to medium-volume roads**, especially under harsh climatic conditions.
viii.	Surface dressing is generally a rough surface with exposed aggregates.	viii.	Otta seal on the other hand is dense, tightly interlocked surface with fewer loose stones.
ix.	It requires more frequent maintenance and resealing.	ix.	It requires less maintenance due to better aggregate interlock.
x.	It is used in maintenance of existing paved roads and low-volume roads.	x.	It is used in rural roads, gravel road upgrading, and roads in developing regions where high-quality aggregate is scarce.

3. **The time for a clay layer to achieve 99% consolidation is 10 years. What time would be required to achieve 99% consolidation if the layer were twice a thick, five times more permeable and three times more compressible? [10]**

Answer:

$$According\ ¿\ Terzaghi's\ one\text{-}dimensional\ consolidation\ theory,$$

$$t=\frac{T_v H_d^2}{C_v}$$

$$where:$$

$$t\ =\ time\ required\ for\ consolidation$$

$$T_v = time\ factor\ (constant\ for\ the\ same\ degree\ of\ consolidation)$$

$$H_d = drainage\ path\ length$$

$$C_v = coefficient\ of\ consolidation$$

$$Also,$$

$$C_v=\frac{k}{m_v \gamma_w}$$

$$where:$$

$$k = \ coefficient \ \ of \ \ permeability$$

$$m_v = coefficient \ \ of \ \ compressibility$$

$$\gamma_w = unit \ \ weight \ \ of \ \ water\,(constant)$$

$$Hence,$$

$$t \propto \frac{H^2 m_v}{k}$$

Given:

- $Initial \ \ time, t_1 = 10 \ years$
- $Thickness \ \ becomes \ \ twice: \ H_2 = 2H_1$
- $Permeability \ \ becomes \ 5 \times: \ k_2 = 5k_1$
- $Compressibility \ \ becomes \ 3 \times: \ m_{v2} = 3m_{v1}$

$$\frac{t_2}{t_1} = \left(\frac{H_2}{H_1}\right)^2 \times \frac{m_{v2}}{m_{v1}} \times \frac{k_1}{k_2}$$

$$Substituting \ \ the \ \ values,$$

$$\frac{t_2}{10} = (2)^2 \times 3 \times \frac{1}{5} 10$$

$$= \frac{t_2}{10} = \frac{12}{5} = 2.4 = 24 \ years$$

$$t_2 = 24 \text{ years}$$

Therefore, the clay layer would require approximately 24 years to achieve 99% consolidation under the given conditions.

4. **Explain why safe disposal of waste water is necessary. Provide comparison between the separate and combined sewer system. [10]**

Answer:

Safe disposal of wastewater is essential to protect public health, the environment, and water resources. If the wastewater is haphazardly discharged without proper treatment, it can cause serious health, environmental, and economic problems. The present condition of Bagmati and Bishnumati rivers in Kathmandu are the major examples of how over-exploitation and direct disposal of wastewater does to the natural resources and environment.

Safe disposal of waste water is necessary because it:

i. **Protects Public Health**
   a. It prevents the spread of waterborne diseases such as cholera, typhoid, dysentery, and hepatitis.
   b. It reduces human exposure to harmful pathogens.

ii. **Prevents Water Pollution**
   a. It stops contamination of rivers, lakes, ponds, and groundwater.
   b. It preserves water quality for drinking, irrigation, and industrial use.

iii. **Protects Aquatic Life**

a. It prevents depletion of dissolved oxygen caused by organic pollutants.
b. It reduces fish kills and protects aquatic ecosystems.

iv. **Prevents Environmental Degradation**
a. It minimizes soil contamination and foul odors.
b. It prevents breeding of flies, mosquitoes, and other disease vectors.

v. **Protects Groundwater Resources**
a. It prevents seepage of untreated wastewater into underground aquifers.
b. It ensures safe groundwater for domestic and agricultural purposes.

vi. **Maintains Public Hygiene and Aesthetics**
a. It keeps communities clean and free from unpleasant smells and unsightly wastewater accumulation.
b. It helps in Improving the quality of the living environment.

vii. **Enables Reuse of Treated Water**
a. Treated wastewater can be safely reused for irrigation, landscaping, industrial cooling, and groundwater recharge.
b. It also conserves freshwater resources.

Separate and combined sewerage systems are the two common types of sewerage systems. The comparison between them is listed in points below:

Separate sewer system	Combined sewer system
1. This sewerage system consists of two different sewers, one for convergence of sewage and another for rainwater.	1. This sewerage system consists of only one sewer to carry both sanitary sewage as well as stormwater.
2. It requires sewer of large area since cleaning is not possible in small sewers.	2. It requires large dimensions of sewer.
3. Load on treatment plant is reduced as waste type is collected separately.	3. Load on treatment plant increases as waste type is collected at same sewer.
4. The system reduces possibility of pollution due to overflow by excess rainfall.	4. Possibility of overflow of sewage during rainy season.
5. Choking problem can be noticed in separate sewer system as self-cleansing velocity is difficult to be achieved.	5. Self-cleansing velocity is easily achieved in combined sewer system.

5. What are the various types of treatment of water to make it safe for drinking? Explain briefly the component of slow sand filter. [10]

Answer:

Water treatment is the process of removing physical, chemical, and biological impurities from raw water to ensure it is safe, potable, and meets quality standards for human consumption. The various types of treatment methods are typically organized into a sequential system, especially when sourcing water from rivers.

i. Physical and preliminary treatments:

a. **Intake Structures and Screening:** The process begins by drawing water through an intake equipped with gates to prevent the entry of large debris. It then passes through **screens** (coarse, medium, or fine) to remove floating matter like sticks, branches, and dead animals, which reduces the load on subsequent units.

b. **Plain Sedimentation:** Water is kept quiescent in a tank for 4 to 8 hours, allowing **coarse suspended particles** such as sand and silt to settle at the bottom by the action of gravity.

c. **Aeration:** Water is brought into contact with air to remove **dissolved gases** like $CO2$ (which reduces its corrosiveness) and to eliminate objectionable tastes or odors.

ii. Chemical treatment:

a. **Sedimentation with Coagulation and Flocculation:** Fine suspended and **colloidal materials** (such as clay) that do not settle easily are treated with chemicals called **coagulants** (e.g., alum, iron salts). These chemicals form insoluble, gelatinous flocs that attract and trap particles, allowing them to be removed through settling.

b. **Water Softening:** This process reduces **temporary or permanent hardness** caused by calcium and magnesium ions. Common methods include:
   a. **Boiling or Adding Lime:** Effective for temporary hardness.
   b. **Lime-Soda Process:** Adds lime and soda ash to form insoluble precipitates that are later filtered out.
   c. **Zeolite or Ionization:** Specialized methods for complete demineralization.

iii. Biological treatment:

a. **Filtration:** Water passes through beds of sand and granular material to remove **small suspended sediments** and impurities whose specific gravity is less than or equal to water.
   - **Slow Sand Filters:** Highly efficient at removing bacterial loads.
   - **Rapid Sand Filters:** Operate at high rates and are effective for color and turbidity removal.

iv. Disinfection:

a. **Disinfection:** This is the most critical stage for making water safe for drinking by **killing pathogenic bacteria and microorganisms**.
b. **Chlorination:** The most common and economical disinfection method. When chlorine is added, it forms **hypochlorous acid (HOCl)**, which penetrates and destroys the cell walls of microorganisms. Different applications include plain, pre-, post-, and break-point chlorination.

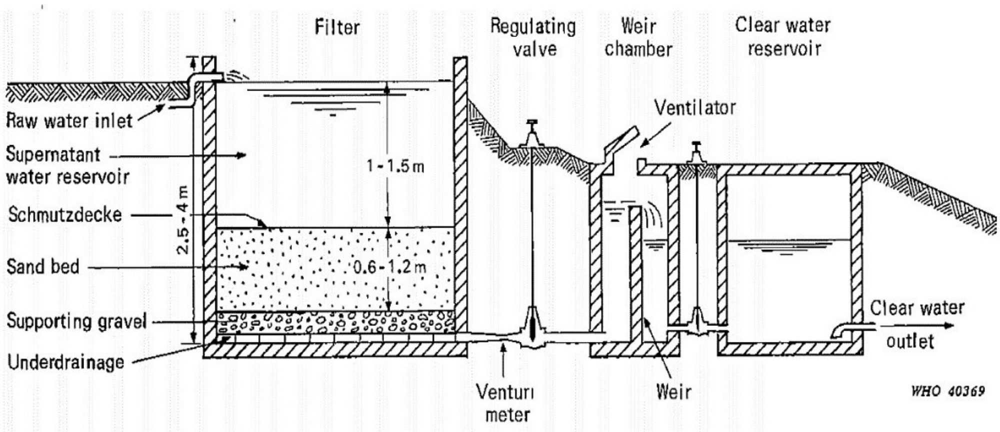
v. Miscellaneous:

Additional processes are utilized to target specific remaining impurities:

a. **Removal of Iron and Manganese:** To prevent staining and metallic taste.
b. **Color and Odor Removal:** Often using activated carbon or additional oxidation.
c. **Arsenic Removal:** Critical for regions where groundwater is contaminated with toxic arsenic.

A slow sand filter (SSF) is a water treatment unit that purifies water by passing it slowly through a bed of fine sand, where suspended and colloidal particles are removed from the water.

The components of Slow sand filter are listed below:



i. Supernatant liquid:

- Water is stored above the sand bed, usually **1.0–1.5 m deep**.
- Provides the hydraulic head required for filtration.
- Allows some suspended particles to settle before filtration.

ii. Filter bed:

- The main filtering medium, consisting of **fine, clean sand**.
- Typical thickness: **0.7–1.0 m**.
- Removes suspended solids, bacteria, and organic matter by straining, adsorption, and biological action.
- A biological layer called the **schmutzdecke** develops on the surface and plays a major role in purification.

iii. Graded gravel layer:

- Located below the sand bed.
- Consists of gravel of increasing size from top to bottom.
- Typical thickness: **0.3–0.5 m**.
- Supports the sand bed and prevents fine sand from entering the under-drain system.

iv. Under-drainage system:

- Made up of perforated pipes or open-jointed drains.
- Collects the filtered water uniformly.
- Prevents loss of filter media and conveys filtered water to the outlet.

v. Filter tank:

- A watertight concrete or masonry structure that contains all filter components.
- Provides structural support and prevents leakage.

vi. Inlet and outlet mechanisms:

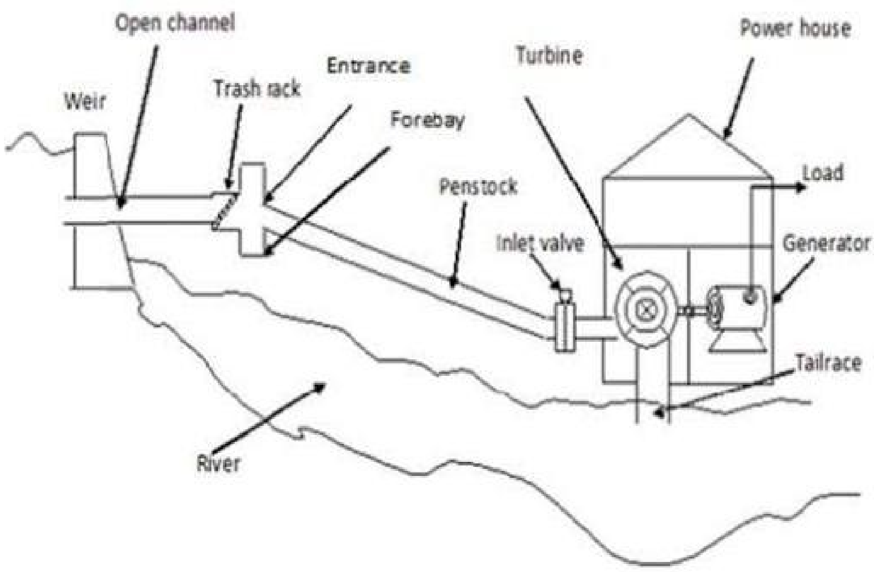
- **Inlet:** Distributes raw water uniformly over the filter surface without disturbing the sand.
- **Outlet:** Collects filtered water and allows control of the filtration rate and water level.

6. Describe about the components of a micro hydro system with neat sketches. [5]

Answer:

A micro hydro system is a small-scale hydropower plant typically **up to 100 kW** that generates electricity by utilizing the energy of flowing water.

The major components and their functions are described below along with the sketch:



i. Intake:

- Intake diverts water from the river or stream into the system.
- It prevents excessive water from entering during floods.

- It often includes a gate to regulate flow.

ii. Trash rack:

- Trash rack is a screen placed at the intake.
- It prevents leaves, branches, stones, and other debris from entering the system.
- It also protects downstream components from damage.

iii. Desilting basin:

- Desilting basin removes sand, silt, and gravel from the diverted water.
- It prevents abrasion and wear of the turbine and other equipment.

iv. Headrace canal:

- Headrace canal conveys water from the intake to the forebay tank.
- It is designed to minimize head loss and maintain a steady flow.

v. Forebay:

- Forebay acts as a small storage and regulating basin.
- It allows remaining sediments to settle.
- It also provides a calm water surface before the penstock.
- Forebay also includes a spillway to discharge excess water safely.

vi. Penstock:

- A pressure pipe that carries water from the forebay to the turbine is known as penstock.
- It converts the available head into high-pressure flow.
- It is usually made of steel, HDPE, or PVC depending on the head and discharge.

vii. Turbine:

- Turbine converts the hydraulic energy of water into mechanical energy.
- The type of turbine depends on the available head and flow.
- Common turbines used in micro hydro system are: **Pelton** (high head), **Cross-flow** (medium head), **Francis** (medium head), and **Kaplan/Propeller** (low head).

viii. Generator:

- Generator is coupled to the turbine shaft.
- It converts mechanical energy into electrical energy.

ix. Control panel:

- Control panel regulates voltage and frequency.
- It protects the system from overloads and electrical faults.
- It also controls power distribution.

x. Tailrace:

- Tailrace canal returns water from the turbine back to the river.
- It ensures smooth discharge without affecting turbine operation.

xi. Transmission and distribution system:

- The system carries the generated electricity to consumers through transmission lines and distribution networks.

7. **Describe with sketches various types of river training works and protection works needed for rivers in the mountains and in the plains. [10]**

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

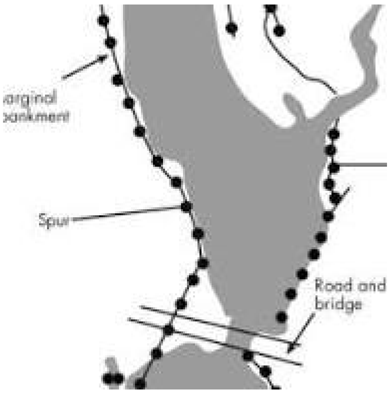
Depending upon the purpose required river training works can be classified into three types.

i. high water training works: training for discharge
ii. low water training works: training for depth
iii. medium water training works: training for sediment

There are different methods of river training works. Some of them are described below:

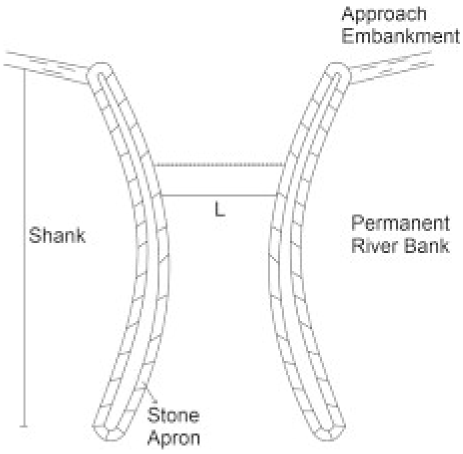
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



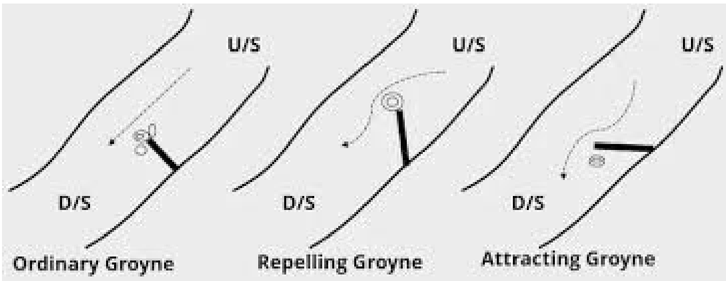
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



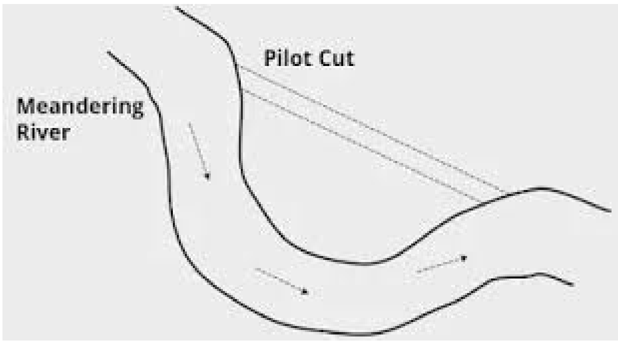
iii. **Groynes or spurs:**

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. **Artificial cutoff:**

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.

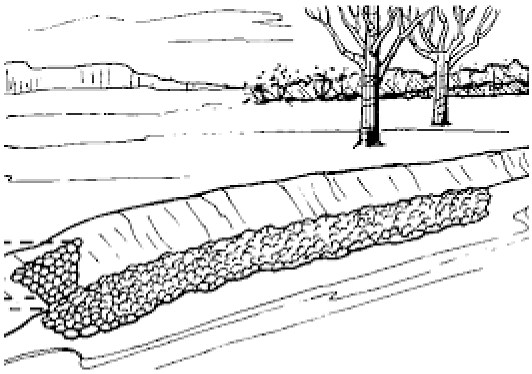


v. **Bank protection and Pitched banks:**

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

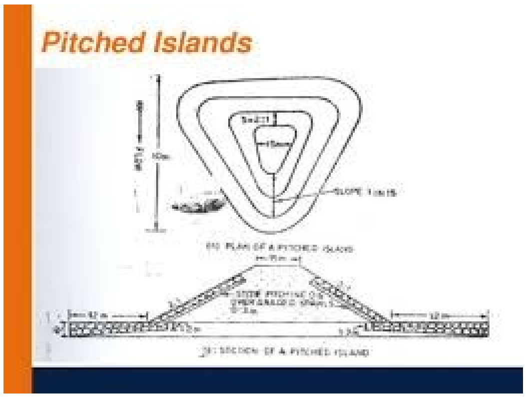
Bank protection work can be classified as direct or indirect.

Direct protection works include work done on the bank itself such as vegetation cover, pitching
Indirect protection includes work constructed not directly on the banks but in front of them.



vi. **Pitched islands:**

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. **Bandalling:**

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.

8. What are the different methods of irrigation? Briefly explain their function and suitability. [10]

Answer:

Irrigation is an artificial application of water to agricultural land in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. **Surface irrigation:**

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.
- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Suitability:
    - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
    - This method is suitable for row and field crops.

  Surface irrigation is further classified into:

  a. Flooding method: entire land surface is covered with water and allowed to stand on the field.
  b. Contour farming: adapted to hilly areas with steep slopes and quickly falling contours.
  c. Furrow irrigation: common method of irrigation where water is applied in small streams between the rows of the crops, grown on the ridges or furrows.

ii. **Sprinkler irrigation:**

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Suitability:
    - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
    - It is suitable for closely spaced crops like cereals and fodders.

iii. **Drip irrigation:**

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- **Suitability:**
    - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
    - The method is ideal for row crops, orchard and vineyards.

9. **Explain why urban planning in Nepal. Describe the challenges being faced in this sector. [10]**

Answer:

Urban planning is the process of planning, designing, and managing the physical, social, economic, and environmental development of towns and cities to ensure orderly, sustainable, and livable urban growth.

Urban planning is important in Nepal due to rapid urbanization**,** population growth**,** rural-to-urban migration, and increasing demand for housing, transportation, water supply, sanitation, and other urban services. Effective urban planning helps improve the quality of life while protecting natural resources and reducing disaster risks.

Urban planning is important in Nepal because of the following reasons:

i. Urban planning promotes planned urban growth and prevents unplanned settlements.
ii. It provides adequate infrastructure, such as roads, water supply, sewerage, drainage, electricity, and solid waste management.
iii. It improves transportation systems and reduces traffic congestion.
iv. Urban planning promotes proper land use through zoning and development control.
v. It helps to protects the environment by conserving open spaces, rivers, forests, and heritage sites.
vi. It reduces disaster risks, especially from earthquakes, floods, and landslides.
vii. It also improves public health through better sanitation and waste management.
viii. It supports economic development by creating efficient commercial and industrial areas.

Challenges faced in urban planning:

i. **Rapid and Unplanned Urbanization**
   - Fast population growth has led to haphazard expansion of cities.

ii. **Inadequate Infrastructure**
   - Shortage of roads, drinking water, drainage, sewerage, public transport, and utilities.

iii. **Poor Land Use Planning**
   - Encroachment on agricultural land, riverbanks, and environmentally sensitive areas.

iv. **Traffic Congestion**
   - Narrow roads, increasing vehicle numbers, and insufficient public transportation.

v. **Solid Waste Management Problems**
   - Inadequate collection, treatment, and disposal of municipal waste.

vi. **Environmental Pollution**
   - Air, water, and noise pollution are increasing due to rapid urban growth.

vii. **Disaster Vulnerability**

- Many urban areas are exposed to earthquakes, floods, landslides, and fires, while building code implementation remains weak.

viii. **Informal Settlements**

- Growth of slums and squatter settlements with limited access to basic services.

ix. **Weak Institutional Capacity**

- Limited technical expertise, financial resources, and coordination among government agencies.

x. **Poor Enforcement of Planning Regulations**

- Building bylaws, zoning regulations, and land-use plans are often not effectively implemented.

xi. **Climate Change Impacts**

- Increased urban flooding, heat stress, and pressure on water resources.

10. a) Describe about the sources of pollution in Nepal. How these sources can be controlled? [5]
b) How can you relate the technology development with environment and society? [5]

Answer:

Pollution is the contamination of the environment by harmful substances or energy that adversely affects human health, ecosystems, and natural resources. In Nepal, rapid urbanization, industrialization, population growth, and poor waste management have increased environmental pollution.

Major sources of pollution in Nepal are described below:

i. Vehicular emissions:

- Rapid increase in the number of vehicles has resulted in generation of harmful emissions resulting in air pollution.
- Use of old and poorly maintained vehicles are mainly responsible for the emissions of smoke, carbon monoxide (CO), nitrogen oxides (NO_x), hydrocarbons (HC), and particulate matter (PM).

ii. Industrial pollution:

The emission of harmful gases and wastages generated from the industries when released in the environment without adequate filtration results in the polluted environment.

- Emissions from brick kilns, cement factories, and manufacturing industries.
- Discharge of untreated industrial wastewater into rivers.
- Release of toxic gases and chemicals.

iii. Domestic waste and sewage:

- Improper disposal of household solid waste.
- Discharge of untreated sewage into rivers and streams.
- Open dumping of garbage.

iv. Agricultural wastages:

- Excessive use of fertilizers and pesticides.
- Livestock waste contaminating nearby water bodies.
- Agricultural runoff polluting rivers and groundwater.

v. Construction and demolition activities:

- Dust generated from road construction, excavation, and building works.
- Noise and vibration from construction machinery.

vi. Natural sources:

- Dust storms, landslides, forest fires, and volcanic ash (rare) also contribute to pollution.

These sources can be controlled by the following ways:

i. Control of vehicular pollution:

- Promote electric and public transportation.
- Regular vehicle emission testing and maintenance.
- Use cleaner fuels and enforce emission standards.

ii. Industrial pollution control:

- Install air pollution control devices (filters, scrubbers, electrostatic precipitators).
- Treat industrial wastewater before discharge.
- Enforce environmental regulations.

iii. Proper solid waste management:

- Segregate waste at source.
- Promote recycling, composting, and sanitary landfills.
- Prohibit open dumping and burning.

iv. Sewage treatment:

- Construct and operate wastewater treatment plants.
- Improve sewerage systems.
- Prevent direct discharge of sewage into rivers.

v. Afforestation and public awareness:

- Increase tree plantation.
- Prevent illegal deforestation and control forest fires.
- Conduct environmental education and awareness programs.
- Strictly enforce environmental laws and pollution standards.

Technological development refers to the advancement and application of scientific knowledge to improve human life. It is closely linked with both the environment and society because technology can improve living standards and economic growth, but it can also create environmental and social challenges if not managed sustainably.

Relationship with society:

i. **Improves Quality of Life**
   a. Provides better healthcare, education, communication, and transportation.
   b. Increases comfort and convenience in daily life.
ii. **Promotes Economic Growth**
   a. Increases industrial productivity and employment opportunities.
   b. Supports innovation and national development.
iii. **Enhances Infrastructure**
   a. Improves roads, bridges, water supply, sanitation, and energy systems.
   b. Supports urban and rural development.
iv. **Improves Communication and Information Access**
   a. Internet and digital technologies connect people globally.
   b. Facilitates education, business, and governance.

Relationship with society:
Positive impacts:

- Development of renewable energy (solar, wind, hydropower).
- Efficient water and wastewater treatment technologies.
- Better solid waste management and recycling.
- Pollution monitoring and environmental conservation.
- Sustainable construction materials and energy-efficient buildings.

Negative aspects:

- Air, water, and soil pollution from industries.
- Deforestation and habitat destruction.
- Excessive consumption of natural resources.
- Greenhouse gas emissions and climate change.
- Generation of electronic (e-waste) and hazardous waste.